

Contents

1 Proof	1
1.1 Statement	1
1.2 Proof (Step by Step)	1
1.2.1 1. Start with the orthogonal set	1
1.2.2 2. Show that b_2 is orthogonal to each a_k	2
1.2.3 3. Show uniqueness of the decomposition	2
1.2.4 4. Final expressions	3
1.3 Summary	3
1.4 Visualisation Insight	3

#proof #ms246 #linear-algebra #chp1

1 Proof

1.1 Statement

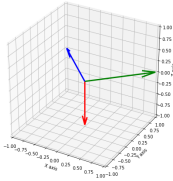
Let $a_1, a_2, \dots, a_m$ be a set of **orthogonal non-zero vectors**, and let b be any other vector with the same number of elements.

Then b can be **uniquely expressed** as:

$$b = b_1 + b_2$$

where:

- b_1 is **dependent** on the set $a_1, a_2, \dots, a_m$,
- b_2 is **orthogonal** to every vector in that set, i.e. $a'_k b_2 = 0$ for all $k = 1, 2, \dots, m$.



1.2 Proof (Step by Step)

1.2.1 1. Start with the orthogonal set

Since $a_1, a_2, \dots, a_m$ are all orthogonal and non-zero, we can write:

$$b = \sum_{j=1}^m \frac{a'_j b}{a'_j a_j} a_j + \left(b - \sum_{j=1}^m \frac{a'_j b}{a'_j a_j} a_j \right)$$

Let:

$$b_1 = \sum_{j=1}^m \frac{a'_j b}{a'_j a_j} a_j, \quad b_2 = b - \sum_{j=1}^m \frac{a'_j b}{a'_j a_j} a_j$$

Hence:

$$b = b_1 + b_2$$

1.2.2 2. Show that b_2 is orthogonal to each a_k

For any $k = 1, 2, \dots, m$:

$$a'_k b_2 = a'_k \left(b - \sum_{j=1}^m \frac{a'_j b}{a'_j a_j} a_j \right)$$

Using orthogonality ($a'_k a_j = 0$ for $k \neq j$):

$$a'_k b_2 = a'_k b - \frac{a'_k b}{a'_k a_k} a'_k a_k = 0$$

Thus, b_2 is orthogonal to **every** a_k .

1.2.3 3. Show uniqueness of the decomposition

Suppose:

$$b = \sum_{i=1}^m c_i a_i + b_2, \quad \text{with } a'_i b_2 = 0 \text{ for all } i.$$

Multiply both sides by a'_j :

$$a'_j b = \sum_{i=1}^m c_i a'_j a_i + a'_j b_2$$

Since the a_i are orthogonal:

$$a'_j b = c_j a'_j a_j$$

Hence:

$$c_j = \frac{a'_j b}{a'_j a_j}$$

Each coefficient c_j is **uniquely determined** by a_j and b .

1.2.4 4. Final expressions

The decomposition is therefore:

$$\begin{aligned} b_1 &= \sum_{j=1}^m \frac{a'_j b}{a'_j a_j} a_j \\ b_2 &= b - \sum_{j=1}^m \frac{a'_j b}{a'_j a_j} a_j \\ b &= b_1 + b_2 \end{aligned}$$

1.3 Summary

- b_1 is the **projection** of b onto the subspace spanned by $a_1, a_2, \dots, a_m$.
 - b_2 is the **component of b orthogonal** to that subspace.
 - Each scalar $c_j = \frac{a'_j b}{a'_j a_j}$ gives how much of a_j contributes to b_1 .
 - The decomposition is **unique**.
-

1.4 Visualisation Insight

Think of $a_1, \dots, a_m$ as the coordinate axes of a smaller “subspace.” The vector b_1 lies *within* this subspace, and b_2 is the “shadow” of b that lies *perpendicular* to it.